

```
mysql> CREATE DATABASE college;  
Query OK, 1 row affected (0.05 sec)
```

```
mysql> USE college;  
Database changed
```

```
mysql> CREATE TABLE student (  
->     id INT PRIMARY KEY,  
->     name VARCHAR(30),  
->     department VARCHAR(30),  
->     age INT,  
->     marks INT  
-> );
```

```
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> INSERT INTO student VALUES (1, 'Ram', 'CSE', 20, 85);  
Query OK, 1 row affected (0.04 sec)
```

```
mysql> INSERT INTO student VALUES (2, 'Shyam', 'ECE', 21, 78);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO student VALUES (3, 'John', 'CSE', 20, 92);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO student VALUES (4, 'Anu', 'EEE', 22, 67);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO student VALUES (5, 'Priya', 'CSE', 21, 88);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO student VALUES (6, 'Kavi', 'ECE', 20, 74);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO student VALUES (7, 'Ravi', 'EEE', 22, 81);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM student ORDER BY marks DESC;
```

id	name	department	age	marks
3	John	CSE	20	92
5	Priya	CSE	21	88
7	Ravi	EEE	22	81
1	Ram	CSE	20	85
6	Kavi	ECE	20	74
2	Shyam	ECE	21	78
4	Anu	EEE	22	67

department	avg_marks
CSE	88.3333
ECE	76.0000
EEE	74.0000

3 rows in set (0.00 sec)

```
mysql> SELECT age, SUM(marks) AS total_marks
-> FROM student
-> GROUP BY age
-> HAVING SUM(marks) > 300;
Empty set (0.00 sec)
```

```
mysql> SELECT department, AVG(marks) AS avg_marks
-> FROM student
-> GROUP BY department
-> ORDER BY avg_marks DESC;
```

department	avg_marks
CSE	88.3333
ECE	76.0000
EEE	74.0000

3 rows in set (0.00 sec)

```
mysql> SELECT department, COUNT(*) AS total_students
-> FROM student
-> GROUP BY department
-> HAVING COUNT(*) > 3
-> ORDER BY total_students DESC;
Empty set (0.00 sec)
```

```
mysql> SELECT department, MAX(marks) AS max_marks  
-> FROM student  
-> GROUP BY department;
```

department	max_marks
CSE	92
ECE	78
EEE	81

3 rows in set (0.00 sec)

```
mysql> SELECT age, SUM(marks) AS total_marks  
-> FROM student  
-> GROUP BY age;
```

age	total_marks
20	251
21	166
22	148

3 rows in set (0.00 sec)

```
mysql> SELECT department, COUNT(*) AS total_students  
-> FROM student  
-> GROUP BY department  
-> HAVING COUNT(*) > 5;
```

Empty set (0.00 sec)

```
mysql> SELECT department, AVG(marks) AS avg_marks  
-> FROM student  
-> GROUP BY department
```



```
mysql> SELECT *  
      -> FROM student  
      -> ORDER BY marks DESC  
      -> LIMIT 5;
```

id	name	department	age	marks
3	John	CSE	20	92
5	Priya	CSE	21	88
1	Ram	CSE	20	85
7	Ravi	EEE	22	81
2	Shyam	ECE	21	78

5 rows in set (0.00 sec)

```
mysql> SELECT department, COUNT(*) AS total_students  
      -> FROM student  
      -> GROUP BY department;
```

department	total_students
CSE	3
ECE	2
EEE	2

3 rows in set (0.00 sec)

```
mysql> SELECT department, AVG(marks) AS avg_marks  
      -> FROM student  
      -> GROUP BY department;
```

department	avg_marks
CSE	88.3333
ECE	76.0000
EEE	74.0000

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM student ORDER BY marks DESC;
```

id	name	department	age	marks
3	John	CSE	20	92
5	Priya	CSE	21	88
1	Ram	CSE	20	85
7	Ravi	EEE	22	81
2	Shyam	ECE	21	78
6	Kavi	ECE	20	74
4	Anu	EEE	22	67

7 rows in set (0.00 sec)

```
mysql> SELECT department, COUNT(*)  
-> FROM student  
-> GROUP BY department;
```

department	COUNT(*)
CSE	3
ECE	2
EEE	2

3 rows in set (0.00 sec)

```
mysql> SELECT *  
-> FROM student  
-> ORDER BY marks DESC;
```

id	name	department	age	marks
3	John	CSE	20	92
5	Priya	CSE	21	88
1	Ram	CSE	20	85
7	Ravi	EEE	22	81
2	Shyam	ECE	21	78
6	Kavi	ECE	20	74